

JEE MAIN REVISION MASTER: LAWS OF MOTION

2026 2 QUESTIONS 3.33% WEIGHTAGE

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 11 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Newton's Laws & Free Body Diagrams	8 PYQs	Dynamics
Moderate	Friction (Static, Limiting & Kinetic)	6 PYQs	Friction

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Newton's Laws & Free Body Diagrams	$\Sigma \vec{F} = m\vec{a}$, $T = \frac{2m_1 m_2}{m_1 + m_2}g$, $a = \frac{m_2 - m_1}{m_1 + m_2}g$
Friction (Static, Limiting & Kinetic)	$f_{\text{max}} = \mu_s N$, $f_k = \mu_k N$, $\theta_{\text{repose}} = \tan^{-1} \mu_s$, $a = g(\sin \theta - \mu_k \cos \theta)$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Newton's Laws & Free Body Diagrams (High Yield — 8 PYQs)

Core Principle: Equations of motion for connected bodies, string-pulley constraints, and apparent weight in non-inertial frames.

■ **Caution / JEE Pitfall:** In an accelerating elevator with upward acceleration a , apparent weight is $N = m(g + a)$, NOT $m(g - a)$.

■ **Speed Trick:** Use string constraint $\Sigma T_i \cdot x_i = 0 \implies \Sigma T_i \cdot a_i = 0$ to relate accelerations in complex pulley systems.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "A mass M is hung with a light inextensible string as shown in figure. Find the tension of the horizontal string...." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "A man weighs 80 kg. He stands on a weighing scale in a lift which is moving upwards with a uniform acceleration of 6m/s^2 . Wha..." [Ans: N/A]
- [2025 Shift] Q#3 (MCQ): "A wooden block of mass M lies on a rough floor. Another wooden block of the same mass is hanging from the point O through strings as shown i..." [Ans: N/A]

3.2 Friction (Static, Limiting & Kinetic) (Moderate Yield — 6 PYQs)

Core Principle: Static self-adjusting friction up to limiting value $\mu_s N$ and constant kinetic friction $\mu_k N$ on rough surfaces.

■ **Caution / JEE Pitfall:** Static friction is self-adjusting: $f_s = F_{\text{(applied)}}$ as long as $F_{\text{(applied)}} \leq \mu_s N$. It does NOT automatically equal $\mu_s N$.

■ **Speed Trick:** In two-block systems, find the maximum common acceleration allowed by friction before determining if slipping occurs.

Representative Source PYQs:

- **[2025 Shift] Q#3 (MCQ):** "A wooden block of mass M lies on a rough floor. Another wooden block of the same mass is hanging from the point O through strings as shown i..." **[Ans: N/A]**
- **[2025 Shift] Q#4 (MCQ):** "A block of certain mass is placed on a rough floor. The coefficients of static and kinetic friction between the block and the floor are $0.4 \dots$ " **[Ans: N/A]**
- **[2024 Shift] Q#1 (MCQ):** "A block of certain mass is placed on a rough inclined plane. The angle between the plane and the horizontal is 30° . The coefficie..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

A mass M is hung with a light inextensible string as shown in figure. Find the tension of the horizontal string.

- (A) $\sqrt{2}Mg$
- (B) $\sqrt{3}Mg$
- (C) Mg
- (D) $3Mg$

Correct Answer: None

Solution Analysis: Let $T_{(1)}$ be the tension in the horizontal string, $T_{(2)}$ be the tension in the slanted string, and $T_{(3)}$ be the tension in the vertical string. The mass M is at rest, so it is in equilibrium. The vertical string supports the weight of the mass. Hence, $T_{(3)} = Mg$. Now consider the knot O . Three tensions act on it: $T_{(1)}$ horizontally, $T_{(2)}$ along the slanted string, and $T_{(3)}$ vertically downward. Since the knot is also in equilibrium, the forces in the vertical direction must balance. $T_{(2)} \sin 45^\circ = T_{(3)} = Mg$. The forces in the horizontal direction must also balance. $T_{(1)} = T_{(2)} \cos 45^\circ = Mg$.

Benchmark Problem 2: KCET 2024 (Q#1)

A block of certain mass is placed on a rough inclined plane. The angle between the plane and the horizontal is 30° . The coefficients of static and kinetic frictions between the block and the inclined plane are 0.6 and 0.5 respectively. Then, the magnitude of the acceleration of the block is [Take $g = 10 \text{ ms}^{-2}$]

- (A) 2 ms^{-2}
- (B) zero
- (C) 0.196 ms^{-2}
- (D) 0.67 ms^{-2}

Correct Answer: None

Solution Analysis: From the diagram, frictional force $f_s = \mu R = \mu mg \cos 30^\circ = 0.6 \times mg \times \frac{\sqrt{3}}{2} = 0.3\sqrt{3}mg = 0.5196mg$. Magnitude of force pulling the block along the plane downward, $F^{\wedge} = mg \sin 30^\circ = mg/2 = 0.5mg$. Since, $f_s > F^{\wedge}$ Hence, block will not move. $\therefore a = 0$